

PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM1 & W2-PM5 | CYBERSECURITY | NETWORKWALKS

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Program/Batch	Networkwalks Cybersecurity & Ethical Hacking Internship (B083)
Date	18 September 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning) W2-PM-Final (Pentest Report)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own local network with Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to mapping live hosts and open services on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in Kali Linux (footprinting) and on a Windows PC with Zenmap installed (scanning). Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance and scanning activities
WHOIS	Find domain registration details (owner, dates, name servers).
WhatWeb	Fingerprint web technologies (server, CMS, plugins, IP).
Nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.

Wafw00f	Detect whether a Web Application Firewall protects the site.
DNSRecon	Enumerate DNS records (NS, MX, SPF, TXT, SRV).
Zenmap (Nmap GUI)	Scan local hosts/subnet to identify open ports and services.

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon. Each tool was used to collect a different type of information about the target.

WHOIS: I ran *whois networkwalks.com* to obtain publicly available domain registration information. The results showed the domain is registered with GoDaddy.com, LLC, created on 6 November 2019 and set to expire on 6 November 2027, with name servers NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM. The registrant contact details were privacy-protected via Domains By Proxy, LLC, and the domain has clientTransferProhibited, clientUpdateProhibited, clientRenewProhibited and clientDeleteProhibited status locks.

WhatWeb: I then used WhatWeb to identify technologies used by the website. The results identified an Apache web server running **WordPress 7.1** with **WordPress Download Manager 3.3.58**, jQuery 3.7.1, Bootstrap 7.1, and the server IP address **192.232.216.135**. The scan also showed the site redirects from HTTP to HTTPS and exposes a Google Tag Manager script and an admin contact email address in the page metadata.

Nslookup: Using nslookup against the Google public DNS resolver (8.8.8.8), I resolved the domain name to its IP address. The result confirmed **192.232.216.135**, matching the IP identified by WhatWeb.

Curl: I used curl with the *-I* option to inspect the HTTP response headers. The response returned HTTP/2 200, an Apache server header, and a *link* header exposing the WordPress REST API endpoint */wp-json/*, along with cookie, caching and referrer-policy headers.

Wafw00f: Next, I used Wafw00f to determine whether a Web Application Firewall was protecting the website. The result identified the site is behind **ModSecurity (SpiderLabs) WAF**.

DNSRecon: Finally, I used DNSRecon to enumerate DNS records for networkwalks.com. The results identified the SOA/NS records (*ns6135.hostgator.com*, *ns6136.hostgator.com*), an MX record pointing to *mail.networkwalks.com* (**192.232.216.135**), an SPF/TXT record authorising *50.87.144.87* and *websitewelcome.com* to send mail, a Google site-verification TXT record, and 8 SRV records for *_autodiscover._tcp* pointing to cPanel's email autodiscovery service.

4.2 Network Scanning with Zenmap

For the second activity, I used Zenmap (the Nmap GUI) to perform network scanning against my own local network. I ran two separate Quick Scans: one against my local subnet, 192.168.56.0/24, and one against the loopback address, 127.0.0.1, to check open ports on my own machine from both a network and a local perspective.

Scan 1 - Local subnet (192.168.56.0/24): Using the command *nmap -T4 -F 192.168.56.0/24*, Zenmap scanned all 256 addresses in the subnet. One host, **192.168.56.1**, was found up (0.00029s latency), with 96 closed TCP ports and 4 open TCP ports: **80/tcp (http)**, **135/tcp (msrpc)**, **139/tcp (netbios-ssn)** and **445/tcp (microsoft-ds)**. The full scan completed in 11.82 seconds.

Scan 2 - Loopback (127.0.0.1): Using the command *nmap -T4 -F 127.0.0.1*, Zenmap scanned the local loopback address. The host **localhost (127.0.0.1)** was found up (0.00035s latency), with 97 closed TCP ports and 3 open TCP ports: **80/tcp (http)**, **135/tcp (msrpc)** and **445/tcp (microsoft-ds)**. This scan completed in 0.29 seconds and did not show port 139, which the subnet scan did report on 192.168.56.1.

After completing the scans, I opened the Topology section in Zenmap and saved the network topology in PDF format. The topology diagram shows my machine ("localhost") connected to two host entries discovered during scanning: localhost (127.0.0.1) and 192.168.56.1, reflecting the two interfaces/addresses scanned above.

Note: Because both scans were run against my own machine (via loopback and via its host-only/VirtualBox adapter address), only a single live host was returned per scan rather than multiple distinct devices. The open ports (HTTP, MSRPC, NetBIOS-SSN and Microsoft-DS/SMB) reflect local services and shares running on my own PC.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified Apache, WordPress 7.1 and WordPress Download Manager 3.3.58, plus JQuery 3.7.1 version and other headers	Attackers may target known vulnerabilities in identified software versions	● Medium
2	Server IP address identifiable	Nslookup resolved networkwalks.com to 192.232.216.135	Provides information about the network location of the web service	● Low
3	HTTP technical information exposed	curl -I returned response headers including a link header exposing the /wp-json/ REST API endpoint	May assist technology fingerprinting and further enumeration	● Low
4	WAF technology identifiable	Wafw00f identified the site is behind ModSecurity (SpiderLabs) WAF	Reveals information about the web application's security architecture	● Low
5	DNS infrastructure information exposed	DNSRecon identified SOA/NS records, an MX record, a TXT/SPF record and 8 SRV (autodiscover) records	DNS information can help build a broader infrastructure profile	● Medium
6	Local network hosts and open services visible	Zenmap quick scans identified HTTP (80), MSRPC (135), NetBIOS-SSN (139, subnet scan only) and Microsoft-DS/SMB (445) open on the scanned host(s)	Exposed SMB/RPC/NetBIOS services are common targets for lateral movement and exploitation if left unpatched	● Medium

Risk level key: ● **Critical** ● **Medium** ● **Low**

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering and host/port discovery. No exploitation or vulnerability validation was performed as part of these two modules.

Therefore, the presence of information such as a software version, IP address, DNS record or open port does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. Review publicly exposed technology information - Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.
2. Keep software updated - WordPress, WordPress Download Manager and other web technologies should be regularly updated and reviewed against current security advisories.

3. Review HTTP headers - HTTP response headers should be reviewed to determine whether unnecessary technical information (such as the /wp-json/ endpoint) is being exposed.
4. Review DNS records regularly - DNS records, including SRV and TXT entries, should be checked periodically to ensure that only required information and services are publicly exposed.
5. Properly configure and monitor the WAF - Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. Close or restrict unnecessary local services - MSRPC, NetBIOS-SSN and Microsoft-DS/SMB should be disabled or firewalled off from untrusted networks unless they are required, since they are common targets for lateral movement.
7. Perform regular internal network discovery - I should periodically scan my own network to identify active devices and open ports.
8. Investigate unknown devices - Any unexpected device discovered during network scanning should be investigated and verified.
9. Maintain network documentation - Network topology and device/port information should be documented and updated regularly.
10. Perform security testing with authorization - Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the networkwalks.com domain. I learned how WHOIS can provide domain registration information, WhatWeb can identify web technologies such as WordPress and its plugins, Nslookup can resolve domain names to IP addresses, Curl can inspect HTTP headers and expose API endpoints, Wafw00f can identify a WAF such as ModSecurity, and DNSRecon can enumerate DNS, mail and service records.

In the network scanning activity, I used Zenmap to run Quick Scans against my local subnet and my loopback address. This identified my machine as a live host on both, discovering open ports for HTTP, MSRPC, NetBIOS-SSN and Microsoft-DS/SMB, and I generated a network topology diagram of the scanned hosts.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network scan responses.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab, against a domain for which permission was already secured and against my own local network.

8. Evidences Collected

1. WHOIS lookup - networkwalks.com

```
2 3 4 |$|
kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-17T09:58:56Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
```

2. WhatWeb scan - networkwalks.com

```
(kali@kali)-[~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US],
HTTPServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/],
UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1], Cookies[__wpdm_client], Country[UNITED STATES][US],
Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client],
IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58],
Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&pvtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy],
UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1], Country[UNITED STATES][US], Email[info@networkwalks.com],
Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58],
Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&pvtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript],
Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
```

3. Nslookup - networkwalks.com

```
(kali@kali)-[~]
$ nslookup networkwalks.com
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```

4. Curl -I HTTP headers - networkwalks.com

```

(kali@kali)-[~]
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com"
" https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token
-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challe
nges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/
v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=sho
rtlink
set-cookie: __wpdm_client=5f2b8ac50173c8ab7b482d62540f4639; path=/; domain=networkwalks.com; secure; Http0
nly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Thu, 17 Sep 2026 10:37:47 GMT
server: Apache

```

5. Wafw00f - networkwalks.com

```

(kali@kali)-[~]
$ wafw00f networkwalks.com

```



```

~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2

```

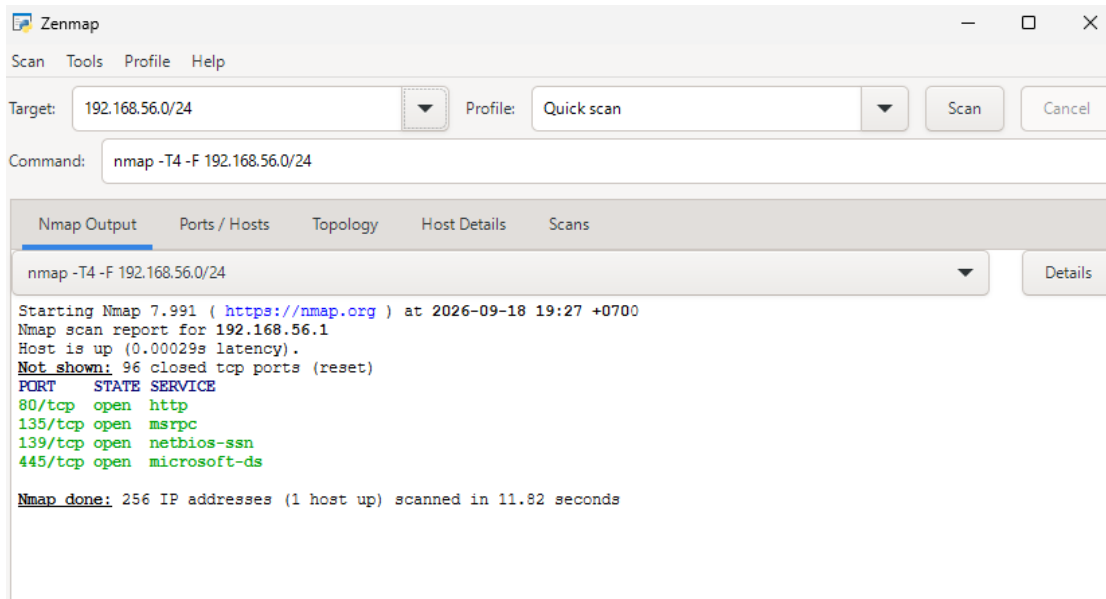
6. DNSRecon - networkwalks.com

```

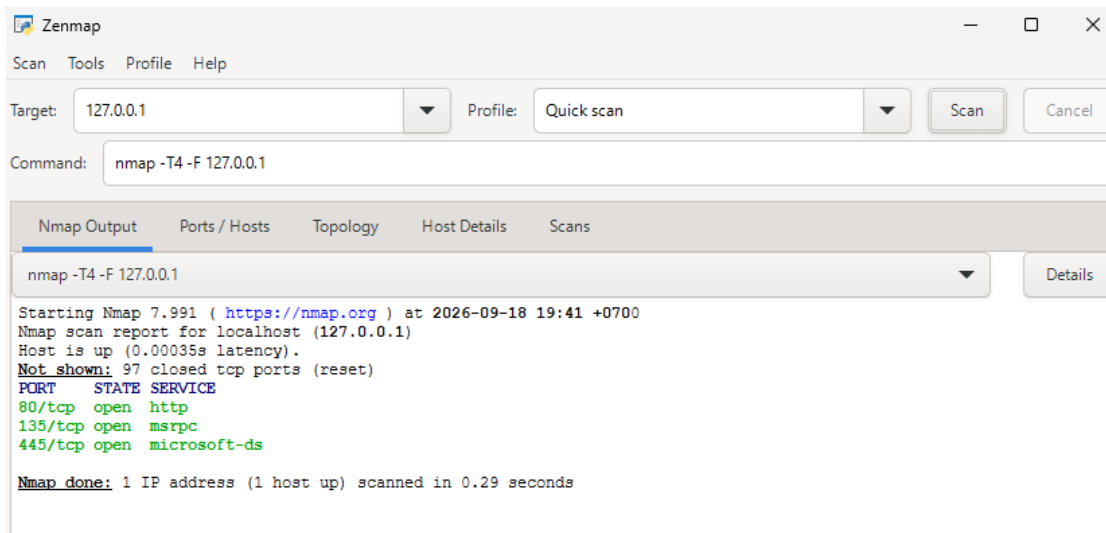
(kali@kali)-[~]
$ dnsrecon -d networkwalks.com
2026-09-17T06:45:17.762538-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-17T06:45:17.763134-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-09-17T06:45:18.195897-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-09-17T06:45:18.491449-0400 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-09-17T06:45:18.898470-0400 INFO NS ns6135.hostgator.com 50.87.144.87
2026-09-17T06:45:19.379833-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-09-17T06:45:19.380266-0400 INFO NS ns6136.hostgator.com 192.232.216.131
2026-09-17T06:45:19.858889-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-09-17T06:45:20.492506-0400 INFO MX mail.networkwalks.com 192.232.216.135
2026-09-17T06:45:20.998628-0400 INFO A networkwalks.com 192.232.216.135
2026-09-17T06:45:21.642291-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-09-17T06:45:21.642471-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqHowY3XemnzDNVK4q75X-Ij-mjgEeg-
UsYI
2026-09-17T06:45:22.061905-0400 INFO Enumerating SRV Records
2026-09-17T06:45:24.676374-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15
443
2026-09-17T06:45:24.676662-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14
443
2026-09-17T06:45:24.677190-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9
443
2026-09-17T06:45:24.677545-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11
443
2026-09-17T06:45:24.677710-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14
443
2026-09-17T06:45:24.678102-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9
443

```

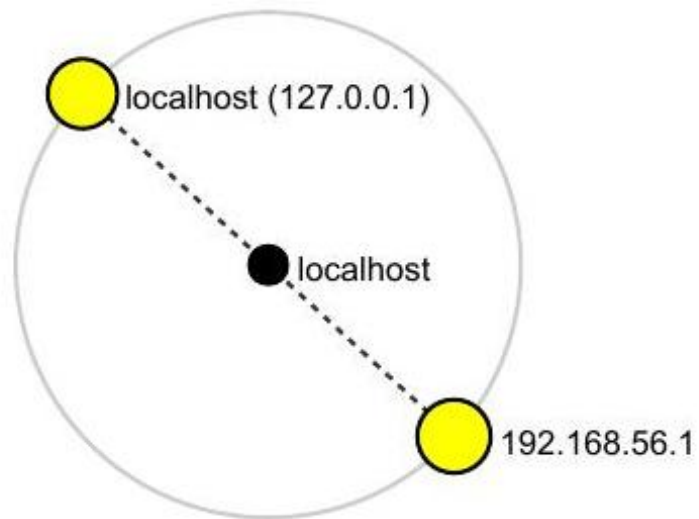
7. Zenmap Quick Scan - 192.168.56.0/24



8. Zenmap Quick Scan - 127.0.0.1 (loopback)



9. Zenmap Network Topology



**-End-
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Project Information

Program Name: Cybersecurity program at Networkwalks | Week: 02 | B083